```c
#include <stdio.h>

int binarysearch(int array[], int key, int n)
{
    int low = 0;
    int high = n - 1;
    int mid;

    while (low <= high)
    {
        mid = (low + high) / 2;

        if (array[mid] == key)
        {
            return mid;
        }
        else if (array[mid] < key)
        {
            low = mid + 1;
        }
        else
        {
            high = mid - 1;
        }
    }

    return -1;
}

int main()
{
    int y, n, key;
    int temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int array[n];

    printf("Enter array elements:\n");

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &array[i]);
    }

    printf("Enter element to search: ");
    scanf("%d", &key);
```

```
    for (int i = 0; i < n - 1; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
            if (array[i] > array[j])
            {
                temp = array[i];
                array[i] = array[j];
                array[j] = temp;
            }
        }
    }

    printf("Sorted array: ");

    for (int i = 0; i < n; i++)
    {
        printf("%d ", array[i]);
    }

    printf("\n");


    y = binarysearch(array, key, n);

    if (y == -1)
    {
        printf("Element not found.\n");
    }
    else
    {
        printf("Element found at index %d.\n", y);
    }

    return 0;
}
```